

1 Autoplay on Trial: An Online Experiment on the Suspect 1
2 Behind Excessive Screen Time 2

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4 July 16, 2025 4

5 **Abstract** 5

6 Interface design can be strategically employed to ‘nudge’ consumers to take certain ac- 6
7 tions and is suspected of promoting addictive online behavior. A prevalent interface 7
8 element across numerous popular social media and streaming platforms is the default 8
9 autoplay feature. In this study, we present an incentivized online experiment to inves- 9
10 tigate whether the autoplay feature alone can cause an unwarranted increase in desired 10
11 viewing times. While controlling for content, we demonstrate that the autoplay feature, 11
12 in isolation, does not override participants’ previously indicated preferences for media 12
13 consumption. We discuss the our findings in terms of what actually renders online plat- 13
14 forms addictive. Our results have direct policy implications for the proposed blanket 14
15 ban on autoplay by the US Congress (SMART Act). 15

*I am grateful to Ernesto Reuben and Kerstin Bongard-Blanchy for their guidance in the design and conception of this experiment, to Sophie Doublet for her help in interface design, to Suvadeep Mukherjee for his guidance in building web experiments. I am also thankful for the invaluable comments from the Behavioral Economics Summer School at the University of Copenhagen, the doctoral workshop at UC Louvain, the Experiments Workshop Soletto at Sapienza University Rome, Samuel Greiff, Manuel Munoz Herrera, Hande Erkut and countless colleagues.

1 Introduction

The rise of social media and streaming platforms have generated discussion around potential societal benefits and concerns about harms such as addiction and self-control issues. A number of recent economic studies have found convincing evidence that people spend significantly more time than they would like to on their smartphones in general (Hoong, 2021) and on social media (Allcott, Gentzkow, & Song, 2022), and social media use was associated with decreased subjective well being (Allcott, Braghieri, Eichmeyer, & Gentzkow, 2020). The reason behind such findings need not be elusive. Online content serving platforms, like Facebook or Netflix, monetize consumer attention and have the ability to control the setting in which consumption decisions occur (e.g., format of information, presentation of choices). Such firms are potentially incentivized to increase the ‘addictiveness’ of their platforms by altering the content and design provided to users, as occupying more of consumers’ free time can lead to higher revenues (Ichihashi & Kim, 2023; Kleinberg, Mullainathan, & Raghavan, 2022). Designing for addiction, in turn, would lead to increased content consumption but decreased utility from consumption, as implied by the extensive literature on present bias (see Ericson and Laibson (2019) for a review). Wary policymakers across both sides of the Atlantic have already made efforts to restrain the negative effects such incentives could have on consumer welfare (see SMART Act by US Congress and Digital Services Act by EU). Both acts are attempts to regulate the increasingly relevant digital environments which contain effective ways to solicit consumers’ attention and influence them to make transactional decisions that may go against their best interests (Lupiáñez-Villanueva et al., 2022). Although both regulators and scientists alike successfully identify the problem, it remains to be seen what specific aspects of such online platforms are problematic, as a blanket regulation over the entire landscape of digital services could offset the many benefits users derive from them. Our study constitutes the first such attempt to our knowledge in disentangling the many interrelated features online platforms have and pinpoints to what specifically requires the attention of the regulators. Relevant to our experimental setting is the

proposed ban on defaults regarding media player controls, where the platform serves content “without an express, separate prompt by the user (such as pushing a button or clicking an icon)”.

Interface designers and computer scientists have already documented the many ways in which interface design can be instrumentalized to ‘nudge’ users in practice (see [Caraban, Karapanos, Gonçalves, and Campos \(2019\)](#) for a review). Nudge theory suggests knowledge about systematic biases in decision-making can be leveraged to support people in making better decisions for themselves ([Thaler & Sunstein, 2009](#)). Much of the existing literature is dedicated to improving decision-making in the context of benevolent government-sponsored nudges (see [Mertens, Herberz, Hahnel, and Brosch \(2022\)](#) for a review). More recently, however, discussions also extended to private sector interventions. This branch of the literature addresses the implications of changing behavior for ‘better’ decisions from the firm’s point of view ([Beggs, 2016](#); [Mills, 2023](#); [Smith, Goldstein, & Johnson, 2013](#)). When firms nudge, there should be a distinction between whether the firm or the recipient of the nudge benefits from it. In other words, nudges can become problematic if the objective of the intervention is to maximize the benefit received by the firm instead of the decision-maker. Media player defaults, such as autoplay, are potentially one such example that could be driving the concerning results attested by literature. If these defaults are conceived to increase welfare of consumers instead, they should be preferred to active decisions or other alternatives. [Carroll, Choi, Laibson, Madrian, and Metrick \(2009\)](#) make a similar comparison between defaults and active decisions in the context of saving decisions, with a focus on modelling decision cost attached to making explicit choices.

The understudied ‘autoplay’ feature, is an interface nudge prevalent across social media and streaming platforms that automatically transitions users to the next video content without providing an explicit opportunity to discontinue consumption. Unless opted-out by users, platforms including Facebook, Netflix, YouTube and TikTok provide autoplay as the *default* media player configuration as of mid-2023. Defaults are effective because they (i) reduce effort, (ii) imply endorsement, and (iii) work hand in hand with

cognitive biases such as loss aversion (Smith et al., 2013). In case of autoplay, the default makes watching successive videos more salient as it removes the opportunity from users to decide whether or not to continue watching. Studies have shown that media player design promotes “mindless” content consumption and reduces user agency (Lukoff et al., 2021). Removing the opportunity to decide whether or not to continue watching leads to increased watching times in children (Hiniker, Heung, Hong, & Kientz, 2018), known to exhibit less self-control than adults. If indeed autoplay promotes digital addiction as suggested by the proposed legislation and the literature, we would expect it to alter the decisions of viewers with self control issues even when proposed content is kept same. We therefore hypothesize that holding content constant, autoplay should result in increased viewing times compared to the control group making active decisions. Further, the concerns of addiction could be verified by contrasting actual watching choices to participants’ previous commitment choices regarding how much they initially wanted to watch videos across conditions. In sum, we conceptualize autoplay as facilitating continuous watching, and compare it in an incentivized experiment to a ‘neutral’ setting where users must actively select videos to watch them.

To determine whether autoplay alters viewers’ past commitment choices, we designed an online experiment where participants spent 20 minutes on a higher-paying real effort task (retyping CAPTCHAs)¹ or a lower-paying leisure task (watching videos). In the treatment condition, videos played automatically, while in the control condition, participants had to click to watch each video. To investigate the effects of autoplay on viewers with heterogeneous self-control issues, we included an additional task to elicit self control issues in our sample. In this task, participants visited the experiment website one day before the 20 minute session and practiced both the real effort task and the leisure task for 2 minutes in a so-called “practice session”. Participants were already randomly assigned to the treatment during the practice session and were able to

¹Charness, Gneezy, and Henderson (2018) categorize CAPTCHAs as a decoding task similar to Augenblick, Niederle, and Sprenger (2015) with the following useful properties: Low difficulty of implementation, small differences regarding the skill required to accomplish the task, and limited learning opportunities to improve at accomplishing the task.

familiarize themselves with the tasks and the media player treatment. After the practice session, participants decided how long they would like to spend on the next day for each task with the help of a slider indicating their payoffs. This decision was reminded to the participants on the second day, but it was not binding for most.² In a separate data collection with a smaller sample, we also measured the demand for a commitment device using a multiple price list to see if participants would pay to turn off the autoplay feature. Our two task experimental setup was inspired by [Kool and Botvinick \(2014\)](#), while the real effort task is akin to [Augenblick et al. \(2015\)](#), and the leisure task resembles [Bonein and Denant-Boémont \(2015\)](#). The contribution of our study is on its novel experimental design which allows testing both established and recent theories on societally relevant topics such as digital addiction. The entire task interface is constructed from scratch with the help of professional designers, and it aims to mimic the user experience on a social media platform. The design brings together extensive possibilities of data collection on behavior with the ease of scaling, while holding internal validity at a high standard. In sum, our design welcomes a plethora of new research questions that can be studied through economic experiments.

2 Design

2.1 Overview

To explore the relationship between default video player settings and self-control, we designed a choice environment that captures the tension between productive effort and immediate gratification. Our experimental setup is centered around two competing tasks: A primary typing task that is tedious and offers limited learning opportunities, and a secondary video-watching task that provides immediate entertainment value. This design mirrors real-world situations where individuals must allocate time between effortful productive activities and readily available leisure alternatives.

²5% of participants who are excluded from the main analysis were bound by the choice they made on the first day to add weight to the decision.

The key feature in our design is the manipulation of autoplay setting in the video interface. Participants can switch between tasks at any time, but the ease of continuing with the leisure activity varies systematically across treatment conditions. This allows us to isolate the causal effect of interface design on time allocation decisions while holding constant individual time preferences and other task characteristics.

2.2 Timeline and Session Structure

We conducted a two-day online experiment to examine how autoplay features affect time allocation between productive and leisure tasks. The multi-day structure allowed us to measure the gap between stated preferences and actual behavior under two different video interface conditions. Participants completed two separate data collection sessions with a cooling-off period of 24 hours (up to 48 hours) between sessions.

Day 1 sessions included a practice session to familiarize participants with the experimental interface, followed by preference elicitation measures. Day 2 featured the main 20-minute work session where participants made actual time allocation decisions. We conducted two waves of data collection during May 2023, as described in Table 1.

Table 1: Summary of experimental design and timeline

	Practice Session	Time Choice	Main Session	Multiple Price List	Decision Horizon
Day 1 (May 10)	✓	✓			24-hour
Day 2 (May 11)			✓		immediate
Day 1 (May 18)	✓	✓		✓	24-hour
Day 2 (May 19)			✓		immediate

Note: This table shows the experimental components administered across two data collection waves, with columns ordered chronologically within each day. The first wave (May 10-11) established baseline treatment effects, while the second wave (May 18-19) incorporated the Multiple Price List treatment to measure demand for commitment devices. Decision horizon refers to the time between preference elicitation and the main experimental session.

139 2.3 Payment Structure and Incentives 139

140 We implemented a performance-based payment system to ensure participants faced 140
141 meaningful trade-offs between tasks. We based payment directly on the seconds par- 141
142 ticipants actively spent on each task, with different compensation rates for the typing 142
143 and video-watching activities. This created clear opportunity costs for time allocation 143
144 decisions while avoiding any completion bonuses that might distort behavior. 144

145 Participants who completed all required elements of the experiment received payment 145
146 within seven days after the second session.³ To maintain data quality, participants who 146
147 failed to meet specified quality criteria for the typing task were not compensated for 147
148 time spent on that task. 148

149 2.4 Technical Implementation and Interface Design 149

150 We designed the experimental platform to provide precise control over the testing envi- 150
151 ronment while maintaining ecological validity. We presented both tasks within a single 151
152 web page using distinct tabs, allowing participants to switch between activities seam- 152
153 lessly while preserving progress. The typing task always served as the default upon 153
154 session initiation, requiring an active choice to switch to the video-watching alternative. 154

155 Our interface incorporated several key features to track behavior precisely. A progress 155
156 bar and timer provided real-time feedback on session duration. We logged all critical 156
157 interactions including tab switches, keystrokes during typing, video controls, and cursor 157
158 movements outside the experimental window. This granular tracking enabled us to 158
159 construct detailed measures of attention and engagement across tasks. 159

160 2.5 Internal Validity Controls 160

161 Online experiments present unique challenges for maintaining experimental control, in- 161
162 cluding participant multitasking, technical disruptions, and environmental variation. We 162

³Prolific policy gives researchers 21 days to transfer funds to participants. The typical delay between study completion and payment is however only a couple of days on average, as indicated on the Prolific website ([link](#)).

implemented comprehensive measures to address these concerns and ensure data quality.

Participant and Technology Screening: We selectively admitted participants using multiple screening criteria to ensure stable experimental conditions. Internet speed requirements filtered out participants with unstable connections (minimum 30 Mbps on day 1, 25 Mbps on day 2). We restricted participation to users with Windows, Linux, or Mac operating systems using Chromium-based browsers exclusively. We excluded Safari and Mozilla Firefox due to restrictive autoplay configurations. Mobile devices and tablets were prohibited to standardize the user experience.

Platform Controls: The experimental website enforced linear navigation, preventing participants from moving backward through the experiment. We disabled right-click functions and keyboard shortcuts. Any attempt to navigate backward reset progress to the landing page. We clearly communicated these restrictions to participants at the experiment’s outset.

Attention and Engagement Monitoring: We implemented real-time tracking of participant attention using JavaScript packages to detect interface visibility disruptions.⁴ This system captured actions such as minimizing the browser, switching between web pages, or launching other applications. The tracking provided second-by-second accountability of participant engagement.

Post-Experiment Validation: After completing the experiment, participants reported any connectivity disruptions and confirmed their genuine engagement with the video-watching task. We combined these self-reports with our objective tracking measures to identify and filter data potentially compromised by distractions or divided attention.

2.6 Tasks

The experiment centers on participants’ time allocation decisions between two distinct activities that differ substantially in their cognitive demands and entertainment value. Both the Typing and Watching Tasks were presented to participants twice: First during

⁴For more information about the package, see [here](#).

the Practice Session to familiarize participants with the interface and task mechanics, and second during the Main Session where performance determined compensation. In both sessions, depending on the randomized assignment to conditions, the key experimental manipulation occurs in the Watching Task.

2.6.1 Typing Task

The Typing Task requires retyping successive CAPTCHAs. CAPTCHAs are computer-generated images that contain random uppercase and lowercase letters as well as randomly placed white spaces.⁵ Each CAPTCHA has a total length of 35 characters, including the white spaces.

We designed the images to be blurred and distorted by lines and dots added on top to increase comprehension difficulty.⁶ The interface always presents these CAPTCHA images with a text box and a submit button. When the submit button is clicked, the next CAPTCHA image appears and the text input for the previous one is stored. The task is identical across all conditions, as illustrated in Figure 1.

We established easy-to-meet quality requirements to ensure participants remained active. These requirements include an overall typing accuracy of at least 70 percent in submitted CAPTCHAs⁷ and a minimum of one CAPTCHA submission per minute spent on the task.

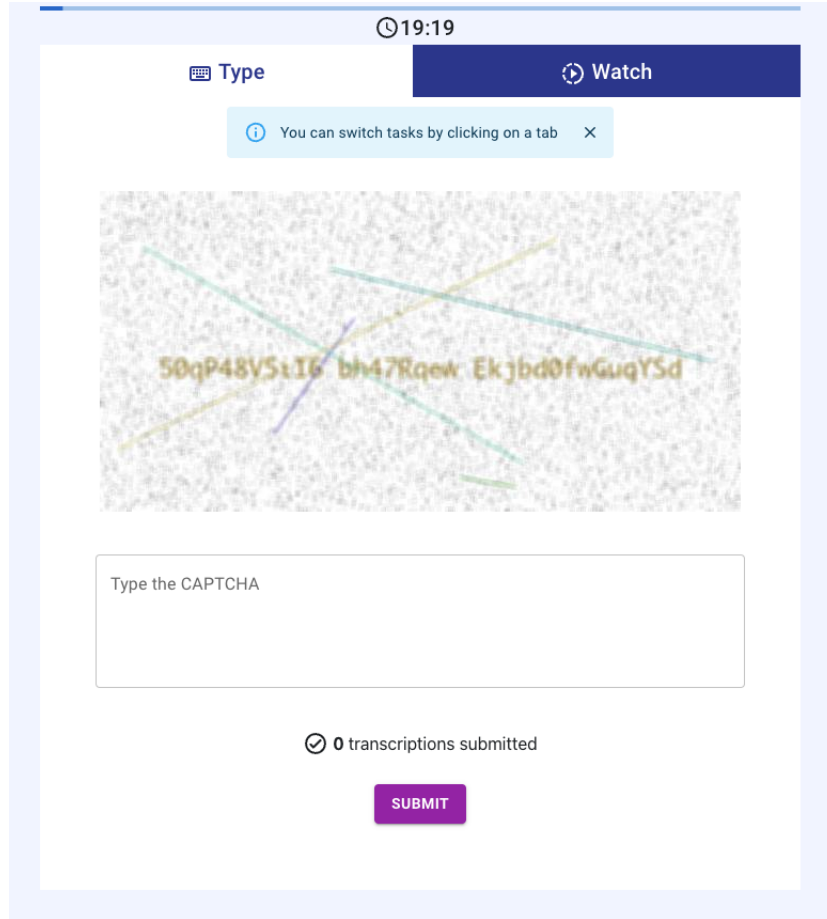
Participants must meet these quality requirements to receive compensation for the Typing Task. We remunerated the Typing Task at 0.38 pennies per second participants worked on it. A hidden timer recorded each second participants kept the Typing Task in the visible part of their screen. Participants in both conditions spent on average 41

⁵White spaces were added to improve readability after pretests. The number of white spaces was limited to a maximum of 5 instances per CAPTCHA to control the typing time per image.

⁶The randomization procedure to generate CAPTCHAs is available [here](#).

⁷We use Python's `difflib` string-matching algorithm to measure accuracy. It consists of finding the longest common substring and then finding recursively the number of matching characters in the non-matching regions on both sides of the longest common substring. Overall accuracy of 70 percent implies an average of 10.5 ($0.3 * 35$) mistakes across all submitted CAPTCHAs.

Figure 1: Typing Task tab



Note: This figure describes the typing interface tab. As participants always begin in the “Type” tab by default, there is a banner that reminds participants to click the tab buttons to switch tasks. This banner disappeared as soon as participants switched tasks once.

213 2.6.2 Watching Task 213

214 In this task, participants watch successive short videos in a customized media player. 214
 215 Since we are interested in isolating the media player’s default settings, the video content 215
 216 and the order in which videos appear to participants are identical across conditions. 216
 217 We completely disabled the media player controls: Participants cannot skip videos (user 217

scrolling is disabled) or use the media player’s slider to advance or go back in a particular video. We remunerated the Watching Task at 0.33 pennies per second participants worked on it.

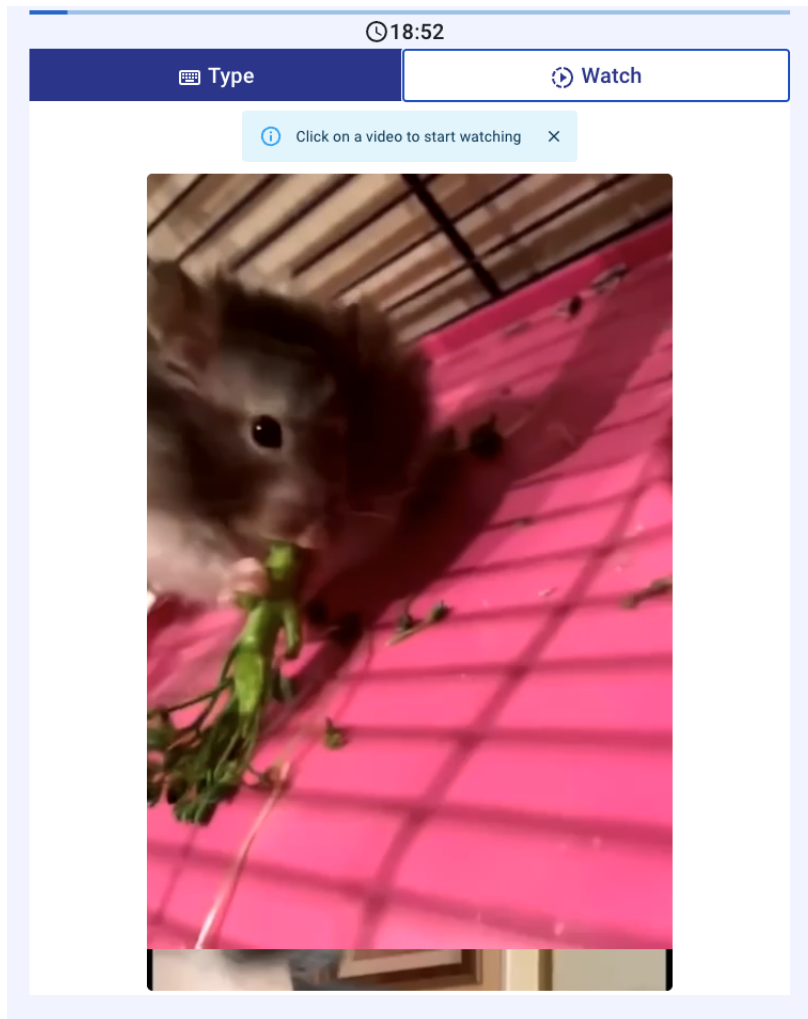
The conditions differ in their autoplay functionality. In the **Autoplay** condition, videos play automatically when the Watching tab is visible and pause with a click anywhere on the video. While the Watching task is visible, the media player keeps scrolling to the next videos and continues playing them unless interrupted. In the **Control** condition, participants need to have the Watching tab visible and click on each consecutive video to play them. At the end of each video, the scrolling animation brings the next video, but the new video only starts playing when the participant clicks on it. Figure 2 illustrates the video tab in the **Control** condition.

We curated videos from YouTube and TikTok, using tags such as “funny animal videos/short” and “cute animal videos/shorts”. We manually selected videos to isolate the difference in the media player design and help avoid algorithmic biases resulting from the recommender system’s personalized suggestions. We chose animal videos because of their popularity during pilot studies.⁸

We manually reviewed the selected videos to ensure that they did not contain any harmful or violent actions toward the animals involved. We kept the videos short to maximize the frequency of experiencing the autoplay feature during the watching session.

⁸During the pilots, the experiment was followed by a separate questionnaire asking participants to rate the videos they had seen and suggest what types of videos they would have liked to see more of.

Figure 2: Watching Task tab



Note: This figure describes the “Watch” tab in the **Control** condition. Before the first video is clicked upon to be played, there is a banner that reminds participants that the video won’t play without clicking. This banner was not present in the **Autoplay** condition and videos started playing as soon as participants clicked on the “Watch” button seen up on the right.

2.7 Practice Session

The experiment consists of two sessions during which participants interact with the tasks described above. The first session is the Practice Session on day one. It comes right after the written descriptions of each task and serves to familiarize participants with

the tasks and the interface. Participants are assigned to either **Autoplay** or **Control** condition before the Practice Session, so that they experience the same video interace on both the Practice and Main Sessions.

During the practice session, each task was available for only 60 seconds and participants were not allowed to switch between tasks. On the first tab, participants were required to retype CAPTCHAs. The 60-second period was enough to submit one CAPTCHA and see the second CAPTCHA image. After 60 seconds, a pop-up appeared on the screen blocking any other interaction and taking participants to the second page.

Participants were required to close this pop-up to advance to the second task. The rationale behind the pop-up was to ensure participants spent equal time on both tasks. Whether or not this pop-up was closed was verified to check participant activity and served as an additional attention check.⁹ Once closed, the Watching Task began for 60 seconds, enabling participants to watch 4 to 5 short videos depending on whether they were assigned to the **Autoplay** or the **Control** condition. In the **Autoplay** condition, videos played continuously, while in the **Control** condition, videos required manual start.

We did not remunerate participants for the time they spent here because the purpose of the practice session was to understand the difficulty and pacing of the tasks, as well as navigation within the interface.

2.8 Multiple Price List

After the Practice Session on day one, participants in the Multiple Price List (**MPL**) condition made 9 binary decisions (see Figure 3). They chose between their preferred media player setting (Autoplay off or on) and a bonus payment. We introduced the MPL condition after the initial data collection phase. We determined its bonuses by the per-second earning differences across tasks, multiplied by the observed time differences between the Autoplay and control groups.

Specifically, the average difference between the Autoplay and control treatments,

⁹All participants closed this pop-up and switched to the Watching Task.

in terms of time spent across tasks, was approximately 60 seconds (as shown in Table 7). This resulted in an average earning difference of $60 * (0.38 - 0.33) = 3$ pennies. To capture this in the MPL choices offered, we set the symmetric bonuses accordingly: $\pounds[0.05, 0.1, 0.25, 0.5]$.

Once these decisions were completed, participants received information on the media player setting for day two—specifically whether Autoplay would be on or off—and the associated bonus. This information was displayed on the next page. With this knowledge, they then decided on their Time Choice. Participants who were not in the MPL condition proceeded directly to the Time Choice page after their Practice Session. More details on the Time Choice can be found in the subsequent section.

Figure 3: Multiple Price List page

Autoplay on or off?

In the practice session videos played automatically with **Autoplay**. There is another version **where you need to click on videos to play them**.

You need to make 9 decisions about Autoplay. One of your decisions will be randomly chosen and implemented. **You will receive a bonus payment and watch videos with the chosen Autoplay setting.**

For example, if you choose Autoplay for the first decision and it is implemented, you will have Autoplay and receive an additional £0.5 bonus payment.

 AUTOPLAY +£0.5	 NO AUTOPLAY +£0
 AUTOPLAY +£0.25	 NO AUTOPLAY +£0
 AUTOPLAY +£0.1	 NO AUTOPLAY +£0
 AUTOPLAY +£0.05	 NO AUTOPLAY +£0
 AUTOPLAY +£0	 NO AUTOPLAY +£0
 AUTOPLAY +£0	 NO AUTOPLAY +£0.05
 AUTOPLAY +£0	 NO AUTOPLAY +£0.1
 AUTOPLAY +£0	 NO AUTOPLAY +£0.25
 AUTOPLAY +£0	 NO AUTOPLAY +£0.5

 Please pick your favorite option for each row.

CONTINUE

Note: This figure describes the MPL page where participants made 9 binary choices between their preferred autoplay setting and bonus payments. Each row presents a choice between “Autoplay” (left) and “No Autoplay” (right) with associated bonus amounts ranging from £0.05 to £0.5. The interface required participants to select one option in each row before proceeding. One of these 9 decisions was randomly selected for implementation in the Main Session, determining both the autoplay setting and any bonus payment for day two.

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2.9 Time Choice

Following the Practice Session, participants transitioned to the Time Choice section.

This section required them to specify their preferred time allocation between the two

281 tasks for the main 20-minute session scheduled for the following day. Participants knew 281
282 the main session would be 20 minutes long and they were free to choose corner solutions, 282
283 even if that meant focusing solely on one task. 283

284 We designed a slider tool with 20-second steps for participants to indicate their time 284
285 allocation preferences. Initially, the slider appeared grayed out and required a click 285
286 to activate. This design aimed to ensure participants considered their choices in an 286
287 active manner. The click-to-activate design had two benefits: It guaranteed participants 287
288 interacted with the slider and helped participants understand the task-related payoff 288
289 structure. 289

290 To emphasize the importance of this decision, we made the choices of 5 percent of 290
291 participants binding in the main session. Binding choices led to payments based on 291
292 the initial Time Choice, but only if participants met the quality standards described 292
293 under Section 2.6.1. We excluded data from these participants from the final dataset 293
294 accordingly. We informed participants whether their allocation was binding with their 294
295 Time Choice reminder on day two, right before the start of the main session. 295

296 The Time Choice served as a benchmark for participants, and participants accord- 296
297 ingly received a textual reminder of it on day two. The Time Choice section offers 297
298 insights into participants' informed time preferences allows gauging their commitment 298
299 to these choices after the 24-hour cooling period. 299

Figure 4: Time Choice page

Time Choice

Now decide how long you would like to spend **tomorrow** on **Typing** and on **Watching Videos**.

Tomorrow you will have **20 minutes** for both tasks.

1 out of every 20 participant will be **bound** by the choice they made today.

Please **click on the slider** and indicate your **Time Choice**:



i You choose to spend **10 minutes 0 seconds** to **Type**.
You choose to spend **10 minutes 0 seconds** to **Watch Videos**.
You would earn **£4.25**.

Tomorrow you will learn whether your **Time Choice** is binding.

CONTINUE

Note: This figure shows the Time Choice interface with a slider tool allowing participants to allocate time between the two experimental tasks for the following day's 20-minute Main Session. The slider initially appears grayed out and requires a click to activate. Real-time feedback shows the selected allocation (here, 10 minutes each) and expected earnings (£4.25). The interface informs participants that 1 out of 20 (5 percent) will be bound by their choice and that they will learn whether their choice is binding on day two.

300 **2.10 Main Session** 300

301 The Main Session took place on day two, approximately 24 hours after the Time Choice. 301
302 We began by reminding participants of their previous day's allocation and informing 302
303 them whether their choice was binding or whether they were free to allocate their time 303
304 as they preferred. 304

305 The session lasted exactly 20 minutes. Participants could freely switch between the 305
306 Typing and Watching Tasks at any time using the tab interface. Unlike the Practice 306
307 Session, there were no time restrictions on individual tasks, and participants had com- 307

plete autonomy over their time allocation decisions. The interface maintained real-time tracking of time spent on each task, and participants could monitor their progress and potential earnings through the progress bar and timer.

We implemented the experimental treatment during this session through the media player’s autoplay functionality. Participants assigned to the **Autoplay** condition experienced automatic video playback when the Watching tab was visible, while those in the **Control** condition required manual clicks to start each video. All other task mechanics remained identical to the Practice Session.

Payment was calculated based on actual time spent on each task, with compensation rates of 0.38 pennies per second for Typing and 0.33 pennies per second for Watching. Participants needed to meet the established quality requirements for the Typing Task to receive compensation for time spent on that activity. The session concluded automatically after 20 minutes, and participants received their final payment calculation before completing the experiment. **ADD AVERAGE EARNING PER PARTICIPANT.**

2.11 Theoretical Framework and Hypotheses

2.11.1 Basic Work-Leisure Allocation Model

We begin with a standard time allocation framework where participants maximize utility over consumption and leisure. Each participant has a fixed time endowment $\bar{T} = 1200$ seconds that must be allocated between work (typing task) and leisure (watching task):

$$\max_a U(wa/p, \bar{T} - a) \quad (1)$$

where a represents time spent working (typing), $\bar{T} - a$ is leisure time (watching), w is the wage rate per second, and p is the price of consumption goods. In our experimental context, we set $w_{typing} = 0.38$ pennies/second and $w_{watching} = 0.33$ pennies/second, creating a clear opportunity cost structure favoring productive work.

Under standard assumptions, the first-order condition yields the familiar result that the marginal rate of substitution between consumption and leisure equals the real wage.

333 However, this framework assumes that all decisions reflect stable, consistent prefer- 333
 334 ences—an assumption we relax through the introduction of dual-system decision-making. 334

335 2.11.2 Dual-System Framework with Content Characteristics 335

336 Following Kleinberg et al. (2022), we model participants as having two decision-making 336
 337 systems that can conflict during leisure consumption. We decompose leisure utility to 337
 338 account for sequential content consumption with potential self-control failures: 338

$$\max_a U\left(\frac{wa}{p}, E[\text{Leisure Utility}]\right) \quad (2)$$

339 where expected leisure utility is characterized by three key parameters: 339

$$E[\text{Leisure Utility}] = \max\left(\frac{v-w}{1-q} - \frac{p \cdot w}{1-p}, 0\right) \quad (3)$$

340 The parameters capture distinct aspects of content: 340

- 341 • **v : Value** — System 2’s intrinsic utility per unit of leisure content 341
- 342 • **q : Span** — Probability that System 2 wants to continue after consuming each 342
 343 content item 343
- 344 • **p : Moreishness** — Probability that System 1 takes automatic control, continuing 344
 345 consumption regardless of System 2’s preferences 345

346 The opportunity cost w represents forgone earnings from not engaging in the typing 346
 347 task. System 2 derives utility $v-w$ during intended leisure consumption, but experiences 347
 348 disutility $-w$ during periods when System 1 maintains control beyond System 2’s desired 348
 349 stopping point. 349

350 2.11.3 Autoplay Treatment Effect 350

351 Our experimental manipulation targets the moreishness parameter through interface 351
 352 design. We implement two conditions: 352

$$\text{Control condition: } p = p_{\text{base}} \quad (4)$$

$$\text{Autoplay condition: } p = p_{\text{base}} + \Delta p \quad (5)$$

The curated video content in our watching task—short, entertaining animal videos from YouTube and TikTok—likely represents content with elevated baseline moreishness p_{base} due to its bite-sized, immediately gratifying nature. The autoplay feature further increases moreishness by reducing opportunities for System 2 to regain control between videos, as noted by Kleinberg et al. (2022): autoplay “decreases the likelihood that system 2 will regain control.”

2.11.4 Two-Period Decision Structure

Our experimental design separates planning from execution across two time periods:

Day 1 (System 2 Planning): Participants make time allocation choices based on expected utility:

$$a_{\text{planned}}^* = \arg \max U \left(\frac{wa}{p}, E[\text{Leisure Utility} | E[p], E[q], E[v]] \right) \quad (6)$$

Day 2 (System 1 Interference): Actual allocation deviates due to realized treatment effects:

$$a_{\text{actual}} = a_{\text{planned}}^* + \text{System 1 deviation}(p_{\text{realized}}) \quad (7)$$

2.11.5 Hypotheses

Building on this theoretical framework, we formulate our preregistered hypotheses:

Hypothesis 1: The autoplay condition will lead to increased time spent watching videos compared to the control condition. This follows directly from $\frac{\partial E[\text{Leisure Time}]}{\partial p} > 0$: higher moreishness increases the expected duration of leisure sessions through more frequent System 1 takeovers.

Hypothesis 2: No significant difference is expected in initial time allocation decisions (Time Choice) between treatment groups. Since treatment assignment occurs

after the Time Choice task, System 2’s planning should be unaffected by the subsequent
 autoplay manipulation: $E[a_{\text{planned}}^*|\text{Autoplay}] = E[a_{\text{planned}}^*|\text{Control}]$.

Hypothesis 3: Participants in the autoplay condition will show greater deviation
 from their initial Time Choice toward watching videos. The increased moreishness should
 amplify System 1 interference, leading to larger deviations from intended allocations:
 $E[|a_{\text{actual}} - a_{\text{planned}}^*||\text{Autoplay}] > E[|a_{\text{actual}} - a_{\text{planned}}^*||\text{Control}]$.

Hypothesis 4: There will be positive demand for a commitment device to disable
 autoplay. Sophisticated System 2 agents should recognize the utility loss from excessive
 moreishness and be willing to pay to avoid it: $E[U|\text{Control}] - E[U|\text{Autoplay}] > 0$,
 generating positive willingness-to-pay for autoplay removal.

These hypotheses collectively test whether interface design features can systemati-
 cally distort optimal time allocation through their effect on the psychological parameters
 governing content consumption, even when participants face clear monetary incentives
 favoring productive work.¹⁰ Our preregistration included hypotheses 1 and 4. We have
 added supporting hypotheses 2 and 3 regarding our expectations about the Time Choice
 section later.

3 Sample and Procedures

The experiment was run entirely online, using widely available open-source tools.¹¹ We
 drew the sample from the Prolific participant pool, an online recruitment platform com-
 monly used for academic research, targeting adult British residents. We further imposed
 the following criteria: Fluency in English, balanced gender representation, and having
 a stable internet connection (> 30 Mbps on the first and > 25 Mbps on the second day
 of the experiment). We did not allow mobile devices and restricted access to the experi-
 ment to Chromium-based browsers to enhance the internal validity of the experiment as
 previously discussed in Section 2.5. Only participants who succeeded on both attention
 checks, reported not having engaged in other activities, and did not have connection

¹⁰To access the preregistered hypotheses, click [here](#).

¹¹We used the [MERN stack](#) to develop and host the website in an entirely free way.

399 issues during the experiment were included in the final dataset. 399

400 For the first stage of data collection, that is to say excluding the MPL treatment, 400

401 we preregistered a two-sample parallel design using our prototype data and tested for 401

402 equality of means across treatments. We found that the required sample size to test 402

403 [Hypothesis 1](#) with the usual parameter values ($\alpha = .05$, $\beta = .2$) would be 114 per treat- 403

404 ment, with an effect size of 65 seconds calculated by the difference in the average time 404

405 spent between conditions ($\mu_{Autoplay} - \mu_{Control}$) and population standard deviation of 165 405

406 seconds. Since attrition in longitudinal experiments is commonplace, we invited 301 par- 406

407 ticipants on the first day. On day two, we invited the same 301 participants and received 407

408 276 complete responses. No participants failed the attention checks more than twice, 408

409 and therefore were not removed from the dataset as per [Prolific policy](#). We dropped 17 409

410 participants who self-reported having connection issues and 24 additional participants 410

411 who stated having engaged in other activities. We also removed 11 participants who 411

412 had their Time Choice enforced randomly. Finally, we removed 40 participants **INSERT** 412

413 **COMPARISON TABLE** that were detected to spend more than 20 consecutive seconds 413

414 away from our experimental platform and the final dataset on which we base the anal- 414

415 ysis consisted of 184 individuals. Table [2](#) shows the demographic characteristics of the 415

416 observed individuals. 416

Table 2: Balance table for the sample

	Autoplay N = 79	Control N = 105	<i>p</i> [A = C]
<i>Gender</i>			
Female	48 (46.15%)	54 (45.00%)	0.510
<i>Age</i>			
Mean (SD)	40.56 (11.66)	40.24 (10.98)	0.725
<i>Employment</i>			
Employed (full time)	62 (59.62%)	67 (55.83%)	0.718
Employed (part time/self)	28 (26.92%)	28 (23.33%)	0.417
Not employed	11 (10.58%)	18 (15.00%)	0.167
<i>Income</i>			
Low (0-15)	28 (26.92%)	34 (28.33%)	0.989
Middle (15-30)	30 (28.85%)	36 (30.00%)	0.505
High (50+)	15 (14.42%)	13 (10.83%)	0.325
<i>Marital Status</i>			
Married/Partnership	56 (53.85%)	60 (50.00%)	0.510
Single	42 (40.38%)	52 (43.33%)	0.534
Previously married	6 (5.77%)	8 (6.67%)	0.927

Note: This table presents demographic characteristics for the sample of 184 participants across treatment conditions. Income categories are presented in thousands of British pounds (£). Employment categories: “Not employed” includes unemployed (looking and not looking) and homemaker. “Previously married” includes divorced, widowed, and separated. *p*-values for binary outcomes are from two-sample tests of proportions; for continuous variables, from two-sample *t*-tests. For employment, part-time and self-employed are together as one category. The sample demonstrates successful randomization with balanced participant characteristics across the conditions.

417 For the MPL treatment, applying the same selection criteria as before, we invited 417

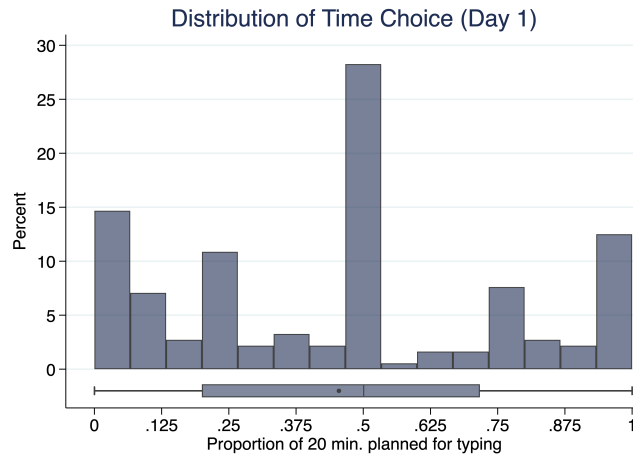
60 participants on day one. On day two, we received 56 complete answers. Once again, no participant failed the attention checks. No participant reported having connection issues. This improvement is likely because we had implemented higher network speed requirements (> 35 Mbps on the first and > 30 Mbps on the second day). We dropped 5 participants who reported having engaged in other activities. As we were interested in the demand for a commitment device, participants did not have their Time Choice enforced in this treatment. We therefore ended up with 51 individuals in the final MPL dataset.

4 Results

4.1 Time Choice: Planned time allocation on day 1

How did participants plan to allocate their time on day 1? We look at the participants' Time Choices to answer this question. These were their incentivized time allocation choices based on how much they would like to spend on the Typing task for the 20-minute main session 24 hours before. Figure 5 illustrates how the Time Choices of participants were distributed along 5 focal points: The modal action for about a third of all participants, was choosing to divide their time equally between the two tasks, then another third chose to only type or watch videos, with 15% choosing typing only and 15% choosing watching only. The remainder of participants were distributed around either $1/4$ of the time typing, or $3/4$ of the time typing. Overall, the day 1 decisions can be characterized as symmetrical around dividing time across either task.

Figure 5: Distribution of Time Choice



Note: This figure shows the distribution of participants' Time Choices for the 20-minute Main Session, measured as the proportion of time planned for typing. The histogram reveals clustering around five focal points: Approximately 20% chose all watching, 12% chose 25% typing, 29% chose equal time allocation, 9% chose 75% typing, and 14% chose all typing. The box plot below shows the overall distribution with median at 0.5.

Based on the observed distribution of the Time Choices, we categorize the day 1 intentions of participants into three distinct groups: Those who almost always planned to type, those who almost always planned to watch videos, and those who planned to mix between the two tasks in various proportions. Table 3 summarizes each group.

Table 3: Participant types by planned time allocation on day 1

	Mostly watching	Alternate	Mostly typing
<i>Time Choice</i>			
Share	28.3%	53.8%	17.9%
Mean (SD)	0.07 (0.08)	0.49 (0.15)	0.95 (0.07)
Median [Min, Max]	0.00 [0.00, 0.23]	0.50 [0.25, 0.75]	1.00 [0.78, 1.00]

Note: This table presents the clustering of participants based on their planned time allocation choices for the 20-minute Main Session on day 1. Watching-focused participants planned to spend 25% or less of their time typing, alternating participants planned to spend between 25% and 75% of their time typing, and typing-focused participants planned to spend more than 75% of their time typing.

442 **4.2 Main Session: Actualized typing decisions on day 2** 442

443 How did participants allocate their time during the 20-minute main session on day 2? 443

444 Because we collected second-by-second data on participant behavior, we can track how 444

445 they allocated their time between the two tasks. Based on the observed distribution of 445

446 the actualized time spent typing, we categorize the day 2 behavior into three groups 446

447 following the same rule: Those who almost always typed (more than 75% typing), those 447

448 who almost always watched videos (less than 25% typing), and those who mixed between 448

449 the two tasks in various proportions. Table 4 summarizes each type. We observe that 449

450 the largest share of participants are now either alternating or mostly typing. 450

Table 4: Participant types by actualized time allocation on day 2

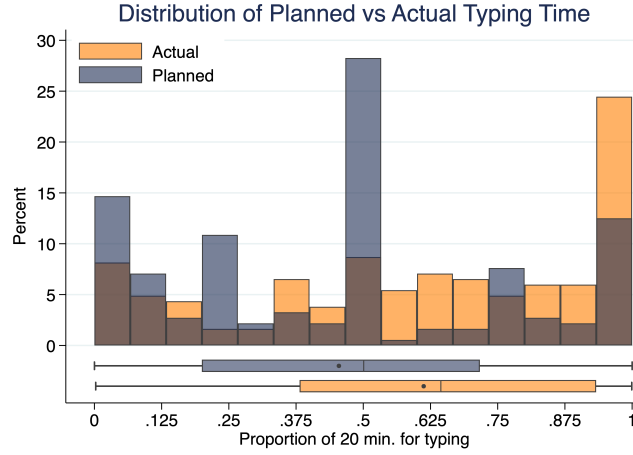
	Mostly watching	Alternate	Mostly typing
<i>Type Time</i>			
Share	19.3%	39.8%	40.9%
Mean (SD)	0.09 (0.08)	0.54 (0.13)	0.94 (0.08)
Median [Min, Max]	0.08 [0.00, 0.24]	0.52 [0.28, 0.74]	0.97 [0.75, 1.00]

Note: This table presents the clustering of participants based on their actualized time allocation during the 20-minute Main Session on day 2. Type Time represents the proportion of time actually spent typing. Mostly watching participants spent 25% or less of their time typing, alternating participants spent between 25% and 75% of their time typing, and mostly typing participants spent more than 75% of their time typing.

4.3 Discrepancy between planned (day 1) and actualized (day 2) time spent on typing

How do participants' day 1 time allocation compare to their actualized time spent on day 2? Participants spent more time on typing than they anticipated, even though we reminded participants of their day 1 plan, and the entire earnings difference between typing for 20 minutes versus watching videos was set to be 60 pence (£0.60). On average, participants planned to allocate 45.5% of their time to typing but actually spent 61.2% of their time on this task, representing an increase of 15.8 percentage points ($t = -8.24$, $p < 0.001$). This extends beyond central tendencies and the entire distribution shifted rightward, with median actual typing time (64.3%) substantially exceeding median planned time (50.0%). The Kolmogorov-Smirnov test confirms that planned and actual distributions differ ($D = 0.86$, $p < 0.001$), indicating systematic deviations from day 1 intentions.

Figure 6: Distribution of Planned vs Actual Time Allocation



Note: This figure compares the distribution of participants' planned time allocation (Day 1) with their actual time spent typing (Day 2), measured as the proportion of the 20-minute session. The rightward shift in the actual distribution demonstrates participants' tendency to spend more time on the effortful typing task than originally planned.

To account for the substantial heterogeneity in both day 1 and day 2 decisions, we compare transitions from participants' day 1 type to their day 2 type in Table 5. We categorize participants into three behavioral types based on their time allocation: mostly watching (< 25% typing), alternate (25-75% typing), and mostly typing (> 75% typing). Overall, 63.0% of participants remained in the same behavioral category between planning and implementation, indicating moderate consistency in behavioral types despite the shift in continuous measures.

Mostly watching and alternate types showed the least adherence to their day 1 plans (59.6% vs 56.6% respectively), with both groups showing upward shifts toward more typing. Among mostly watching planners, 40.4% moved to higher typing categories (30.8% to alternate, 9.6% to mostly typing). Similarly, 40.4% of alternate planners shifted to mostly typing, while only 3.0% moved down to mostly watching. In contrast, mostly typing planners exhibited the highest stability at 87.9%. A chi-square test confirms that deviation rates differ significantly across initial behavioral types ($\chi^2 = 109.00$,

$p < 0.001$), indicating that initial behavioral intentions are strong but heterogeneous predictors of subsequent implementation patterns.

Table 5: Transition matrix: Planned vs actual behavior types

Actual (Day 2)	Planned (Day 1)		
	Mostly watching	Alternate	Mostly typing
Mostly watching	59.6%	3.0%	3.0%
Alternate	30.8%	56.6%	9.1%
Mostly typing	9.6%	40.4%	87.9%

Note: This table shows transitions between planned behavior types (Day 1) and actual behavior types (Day 2). Mostly watching (< 25% typing time), Alternate (25-75% typing time), Mostly typing (> 75% typing time). Diagonal entries show participants who remained consistent (63.0% overall). Off-diagonal entries show deviations: mostly watching planners (40.4% deviated), alternate planners (43.4% deviated), and mostly typing planners (12.1% deviated).

This represents a departure from traditional models of time inconsistency and present bias, which typically predict that individuals will succumb to immediate temptation rather than adhere to effortful plans. Instead, participants demonstrated what might be characterized as ‘reverse present bias’ and systematically allocated more time to the cognitively demanding task than originally intended. The robustness of this finding across the distribution suggests that factors beyond self-control failures govern planning and its implementation in our setting, and potentially include how both tasks and the decision environment were perceived in the experiment.

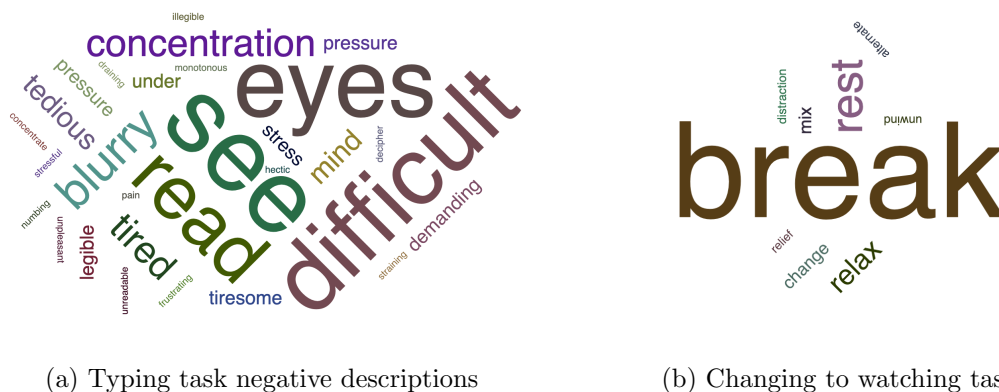
4.4 Experimenter demand effects

Why did participants end up typing more than their day 1 plans? To bridge the gap between our theoretical view of the experimental setup and how participants perceived it, we asked how participants decided to allocate their time during the 20-minute main session once the session was concluded. We explore these by analyzing participants’

493 qualitative answers to an open-ended question on how they decided to divide their time 493
 494 between the two tasks using dictionary-based concept detection methods reviewed by 494
 495 [Ash and Hansen \(2023\)](#). 495

496 We employ dictionary-based concept detection methods within a bag-of-words frame- 496
 497 work to analyze participants’ qualitative responses. Following standard approaches in 497
 498 computational text analysis, we preprocess the text data by converting to lowercase and 498
 499 apply pattern matching to identify specific term sets that relate to negative perceptions 499
 500 of the typing task and breaks towards the watching task. Using regex pattern matching, 500
 501 we create binary indicators for documents containing predetermined words associated 501
 502 with task difficulty (e.g., “tiresome”, “frustrating”, “difficult”) and break-taking behavior 502
 503 (e.g., “break”, “rest”, “relax”) - with the exception of “switch” which typically described 503
 504 simple task changing behavior without the implication of taking a break from typing. 504
 505 This approach allows quantifying the prevalence of common conceptual themes across 505
 506 participants’ obligatory open-ended responses. We match algorithmic output with a 506
 507 manual review to confirm our results about participant perceptions. Figures 7a and 7b 507
 508 summarize these words by their frequency. 508

Figure 7: Word clouds for participant perceptions



Note: Panel (a) shows words describing negative perceptions of the typing task. Panel (b) shows words describing switching to the watching task as break-taking behavior. Word size corresponds to frequency of mention across responses.

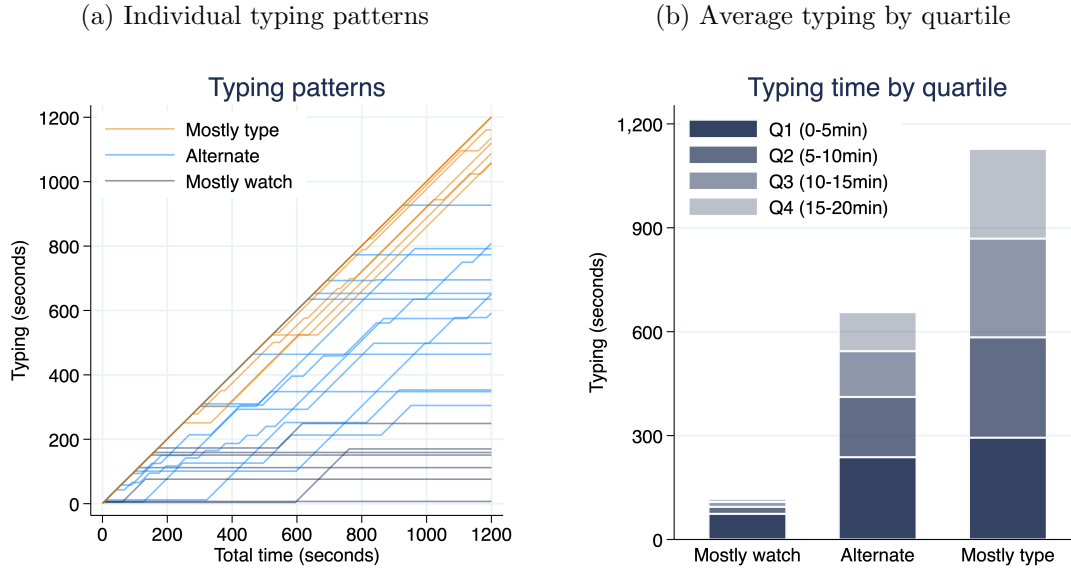
Our mixed methods analysis reveals that when asked to describe how they decided to switch between the tasks, 29% of participants described the typing task negatively at their own will, i.e., without being asked to comment on the typing task itself, while 23% specifically described watching videos as a break from the typing task. These findings suggest that the decision environment was perceived by many as a work environment, where one should type as much as possible while taking as few breaks as possible. The observed increases across all but always typing types, along with the qualitative perceptions suggest that experimenter demand effects have effected participant behavior.

4.5 Effect of achieving day 1 typing objective

One test for this hypothesis is looking at what participants do around the window of achieving their day 1 goals. If their behavior changes, i.e., they start typing less after achieving their goal, this would indicate adherence to day 1 goals and limited distortion due to experimenter demand effects. Should they continue typing as if nothing happened, we should be worried of severe problems in the desing of the experiment. We test this exploratory hypothesis in the next section.

We categorize participants in their actualized typing time on day 2 using the same three categories from day 1 (see Figure 8a). For participants who are “alternating”, typing periods are interrupted by watching periods, following with the “taking breaks” analogy made by participants. We observe substantial individual heterogeneity in lengths of typing and watching periods, and divide the main session across 5-minute periods to understand typing to watching proportions of participants. We find that typing periods get shorter and shorter, suggesting changing between tasks may be strategic as participants get tired of typing or achieve their day 1 goals (see Figure 8b).

Figure 8: Time spent typing



Note: Panel (a) shows typing time for 30 random participants across the 20-minute session. Panel (b) shows average typing time by 5-minute periods, revealing declining typing duration over time.

Why do participants reduce time spent typing as they advance? If they are typing without regard to their day 1 objective, their typing shouldn't be affected by attaining their day 1 goal, and reduce gradually as fatigue or frustration grows from doing the typing task. Instead, if they care about hitting their goal, we should see a sharp drop in time spent typing right after attaining it.

To examine behavioral changes as participants achieve their day 1 plans, we implement a sharp regression discontinuity design using the moment participants reach their planned typing duration as the cutoff. We aggregate binary task choices (whether typing or watching) into 30-, 60-, and 90-second bins around this threshold and estimate the discontinuous change in the probability of choosing the typing task. Results in Table 6 reveal a significant negative effect immediately after plan achievement: participants are 27.4 percentage points less likely to continue typing in the 30-second window following plan completion (robust estimate: -0.274 , $p = 0.007$) and 25.5 percentage points less

likely in the 60-second window (robust estimate: -0.255, $p = 0.003$). However, this effect dissipates over longer horizons, becoming statistically insignificant in the 90-second window (robust estimate: -0.075, $p = 0.487$). These findings suggest that achieving initial plans triggers an immediate behavioral shift toward leisure consumption, consistent with participants treating plan completion as a license to reduce effort temporarily.

Table 6: Regression Discontinuity estimates on typing after achieving day 1 plans

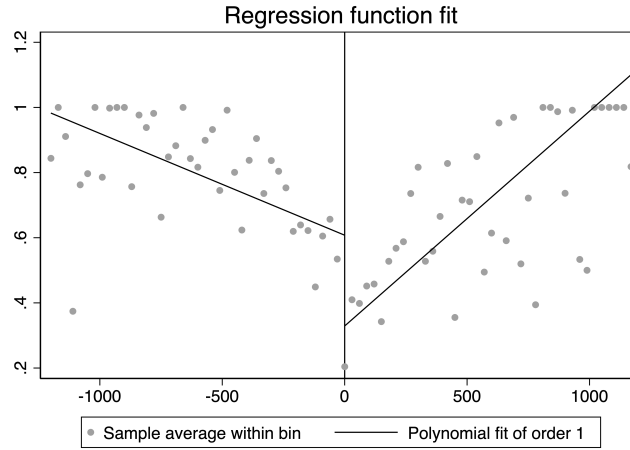
	Time Window		
	30-seconds	60-seconds	90-seconds
Proportion typing	-0.274*** (0.101)	-0.255*** (0.087)	-0.075 (0.108)
Obs. (left)	66,786	67,985	67,119
Obs. (right)	80,270	89,006	92,999
Clusters (left)	157	154	155
Clusters (right)	154	157	161
Bandwidth	339.9	368.3	422.3

Note: This table presents robust regression discontinuity estimates of the treatment effect on task choice at the moment participants achieve their day 1 planned typing duration. Standard errors clustered by participant in parentheses. The outcome variable is a binary indicator for choosing the typing task. Running variables are binned at 30-, 60-, and 90-second intervals around the achievement threshold. All estimates use triangular kernel with MSE-optimal bandwidth selection. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Visual inspection along with the results from 90-second window suggests this discontinuity it not all there is to know about participant behavior: While there is an initial drop in typing rate after achieving day 1 plans, participants eventually ramp up the time they spend at typing (see Figure 9). The positive slope on the right hand side illustrates this effect. Participants who achieve their day 1 plans earlier may in fact eventually get

“bored” of the watching task as well - a behavior that shows how our efforts to create an addictive leisure task may have ultimately failed. These observations are coming from the about at least a third of all participants who were mostly watchers or alternaters in their day 1 types, but have increased their typing task consumption on day 2 (about respectively $(184 * 0.28 * 0.4)/184 = 11,2\%$ and $(184 * 0.54 * 0.4)/184 = 21,6\%$ of participants).

Figure 9: Linear fit for typing rate before and after achieving day 1 plan



Note: This figure plots the average typing rate across 30-second bins for participants as they approach their day 1 goal. The negative slope on the left hand side indicates participants working less and less as they approach the cutoff. The drop showcases the discontinuity where achieving the day 1 goal causes in typing behavior. The positive slope on the right hand side reveals another trend: As time goes on, participants tend to go back to the typing task and decrease their video consumption. This demonstrates participants’ tendency to spend more time on the effortful typing task than originally planned.

4.6 Does Autoplay increase time spent watcing videos?

As participants were mostly focused on typing, they did paid less attention to the watching task, i.e, reducing its consumption value as evidenced by day 2 behavior. Consequently, in our setting experimenter demand effects is acting as a confound, in the sense

of making more difficult to show statistically significant evidence in support of the hypotheses regarding the direction of deviations from the day 1 plans and the effects of the **Autoplay** condition. A lack of evidence in support of these objectives in our setting could be due not to a genuine failure of the corresponding hypotheses but rather to experimenter demand effects [Zizzo \(2010\)](#). With these in mind, we summarize variables of interest from the main 20-minute session across both conditions in Table 7. Two sample t -tests show no significant differences in any of the outcome variables. This suggests that **Autoplay** condition had no impact on the watching behavior observed in the experiment. Further, as established before, participants spend more time typing when compared to their Time Choice on day 1 ($0.158, t(183) = 8.92, p < 0.001$).

Table 7: Sample statistics for variables of interest

	Autoplay	Control	<i>p</i>
	N = 79	N = 105	[A = C]
<i>Time Choice</i>			
Mean (SD)	0.48 (0.31)	0.44 (0.33)	0.410
Median [Min, Max]	0.50 [0.00, 1.00]	0.50 [0.00, 1.00]	
<i>Proportion Typing</i>			
Mean (SD)	0.65 (0.31)	0.58 (0.33)	0.173
Median [Min, Max]	0.72 [0.00, 1.00]	0.56 [0.00, 1.00]	
<i>Nb. of sessions</i>			
Mean (SD)	5.76 (4.98)	4.89 (5.09)	0.246
Median [Min, Max]	4.00 [1.00, 18.00]	3.00 [1.00, 28.00]	
<i>Session length</i>			
Mean (SD)	456.11 (383.22)	546.35 (411.53)	0.131
Median [Min, Max]	314.75 [68.17, 1212.00]	418.67 [44.93, 1314.00]	
<i>Nb. of retyping submissions</i>			
Mean (SD)	19.76 (12.11)	17.21 (11.62)	0.150
Median [Min, Max]	19.00 [0.00, 53.00]	15.00 [0.00, 45.00]	
<i>Nb. of videos watched</i>			
Mean (SD)	31.33 (26.34)	32.72 (26.39)	0.723
Median [Min, Max]	25.00 [0.00, 80.00]	34.00 [0.00, 77.00]	
<i>Content rating</i>			
Mean (SD)	6.96 (2.18)	6.62 (2.45)	0.327
Median [Min, Max]	7.00 [2.00, 10.00]	7.00 [0.00, 10.00]	

Note: This table presents key outcome variables from the Main Session. “Time Choice” refers to participants’ stated preferences for typing on day 1 (proportion), while “Proportion Typing” represents the actual proportion of time spent on typing task during the 20-minute Main Session on day 2. Nb. of sessions is the number of typing or watching sessions participants completed and Session length represents the mean duration of individual sessions in seconds. Content rating reflects participants’ evaluation of video content on a 0-10 scale. *p*-values are from two-sample *t*-tests comparing experimental conditions.

575 To formally test the effects of the **Autoplay** condition on time spent on typing, we 575
576 run the following regression: 576

$$Y_i = \alpha + \beta \text{Autoplay}_i + \gamma \text{TimeChoice}_i + \delta(\text{Autoplay}_i \times \text{TimeChoice}_i) + \epsilon_i \quad (8)$$

577 where Y_i represents our outcome variables of interest (proportion of time spent typing or 577
578 number of videos watched), Autoplay_i is a binary indicator for the autoplay treatment 578
579 condition, and TimeChoice_i is the standardized proportion of time participants planned 579
580 to spend typing on day 1. We estimate this model with and without the interaction term 580
581 to examine whether the autoplay effect varies based on initial planning preferences. 581

582 Our regression analysis reveals that the autoplay manipulation had no statistically 582
583 significant effect on actual typing behavior. Table 8 presents results for typing proportion 583
584 as the dependent variable. The coefficient on the autoplay treatment ranges from 0.12 584
585 to 0.20 across specifications but remains statistically insignificant in all models (p-values 585
586 > 0.16). However, participants' day 1 plans emerge as a powerful predictor of day 2 586
587 behavior: a one standard deviation increase in planned typing time is associated with 587
588 a 0.67 standard deviation increase in actual typing time ($p < 0.001$). Including day 1 588
589 plans increases the model's explanatory power dramatically, with R-squared rising from 589
590 1% to 46%. 590

Table 8: Effect of Autoplay Treatment on Typing Behavior

	Standardized Typing Proportion		
	(1)	(2)	(3)
Autoplay	0.203 (0.147)	0.121 (0.109)	0.119 (0.110)
Time Choice (std.)		0.673*** (0.057)	0.651*** (0.081)
Autoplay $\times$ Time Choice			0.057 (0.110)
Constant	-0.087 (0.100)	-0.052 (0.076)	-0.053 (0.076)
Obs.	184	184	184
R-squared	0.010	0.462	0.463
F-statistic	1.91	77.42	61.52

Note: Robust standard errors in parentheses. All variables are standardized. Time Choice represents participants' planned typing proportion from day 1. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Similarly, Table 9 examines the effect on video consumption, directly testing our primary hypothesis that autoplay would increase video watching. Across all specifications, the autoplay coefficient is statistically insignificant and even negative in the baseline model (-0.053 , $p = 0.723$). This finding contradicts our theoretical prediction and suggests that the autoplay feature did not meaningfully alter participants' video consumption patterns. Again, day 1 plans prove to be the dominant predictor: participants who planned more typing watched significantly fewer videos ($\beta = -0.66$, $p < 0.001$), explaining 44% of the variance in video consumption.

Table 9: Effect of Autoplay Treatment on Video Watching Behavior

	Standardized Videos Watched		
	(1)	(2)	(3)
Autoplay	-0.053 (0.149)	0.029 (0.112)	0.031 (0.112)
Time Choice (std.)		-0.664*** (0.056)	-0.619*** (0.079)
Autoplay $\times$ Time Choice			-0.114 (0.105)
Constant	0.023 (0.098)	-0.012 (0.076)	-0.010 (0.076)
Obs.	184	184	184
R-squared	0.001	0.440	0.443
F-statistic	0.13	73.06	61.53

Note: Robust standard errors in parentheses. All variables are standardized. Time Choice represents participants' planned typing proportion from day 1. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

To summarize, we observe two key patterns in the data that help explain our null findings. First, given the actual effect sizes observed, our power calculations based on pretests were insufficient: The required sample size to provide a two-sided test for Hypothesis 1 with the usual parameter values ($\alpha = .05$, $\beta = .2$) would be 600 per treatment, with an effect size of 0.12 and common standard deviation of 0.739.¹² Other solutions to this problem are extending the main-session duration to above 20 minutes, or coming up with a design that nudges participants to mix more between tasks (i.e., higher earnings when mixing). Second, we observe seemingly contradictory results: the autoplay condition increases time spent typing by 0.12 standard deviations (though statistically insignificant) while simultaneously increasing the number of videos watched

¹²Taken as the root MSE from the standardized proportion typing variable in the third regression.

by 0.03 standard deviations (also insignificant). If people type more, shouldn't they watch fewer videos? We identified a mechanical confound in our experimental design: the total time lost during video transitions. Participants in the **Control** condition, who had to manually click to start each new video, lost an average of 46.52 seconds due to these transition delays whereas those in the **Autoplay** condition lost only 1.08 seconds. This large difference of 45.45 seconds per session is statistically significant ($t(182) = 4.25, p < 0.001$). We find that this difference in time lost not watching videos plausibly explain the contradicting results and underscores the importance of looking at the correct outcome variable (i.e., number of videos watched) when evaluating the effect of the autoplay feature.

Table 10: Total Transition Time by Treatment Condition

	Control	Autoplay	Difference
Total time lost	46.52 (9.27)	1.08 (0.65)	45.45*** (9.29)
Observations	105	79	184
t -statistic			4.89

Note: Standard errors are in parentheses. Time is measured in total seconds lost to video transitions per session. The difference was tested using a two-sample t-test with unequal variances. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

4.7 Effects of Autoplay on deviations from Time Choice (H3)

We then move on to the [Hypothesis 3](#), which suggests participants in the Autoplay condition would deviate more towards watching videos from their initial Time Choice. To assess this hypothesis, we transformed the Time Choice variable by normalizing to percentages and calculated the deviation by subtracting the percentage of actual time spent on the Typing Task from it. Figure 10 shows the distribution of deviations from Time Choice across treatments which appears left-skewed, suggesting increases in

actual typing instead. Specifically, negative values in the figure represent increases in typing, whereas positive values are increases in watching. Applying the Mann-Whitney U test, our data ($U = 6134, p = 0.827$) did not support a significant deviation for the Autoplay condition towards watching. Further descriptive statistics reveal a mean deviation of -11.73 , a median of -8.46 , and a mode of 0, emphasizing the observed left skew. To strengthen our findings, we used the Wilcoxon Signed-Rank test to compare the distribution of deviations to a 0 median distribution, and rejected the null ($W = 5152, p < 0.001$). In summary, contrary to our hypothesis, participants generally spent more time on the Typing Task than they initially intended regardless of treatment.

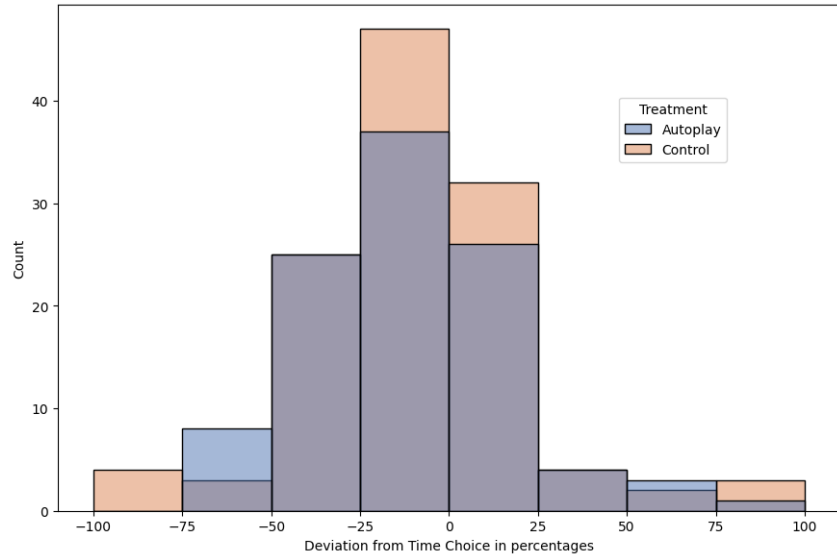


Figure 10: Distribution of the deviation from initial Time Choice in percentages across treatments

4.7.1 Demand for the commitment device (H4)

[Hypothesis 4](#) suggests participants would choose to turn Autoplay off when given the possibility, meaning there is positive demand for the commitment device. We assess this demand by comparing the share of participants choosing to turn Autoplay on or off along with the associated bonus payments for each of the nine decisions in the price list.

Figure 11 shows the percentages of participants preferring Autoplay in blue and those preferring to turn Autoplay off in orange. Distributions clearly show that when given a choice, participants have a preference for Autoplay. Specifically, more than 75 percent of participants chose Autoplay without bonus, 12 percent chose Autoplay if it meant giving up £0.05, and 7 percent chose Autoplay if it meant giving up £0.1.

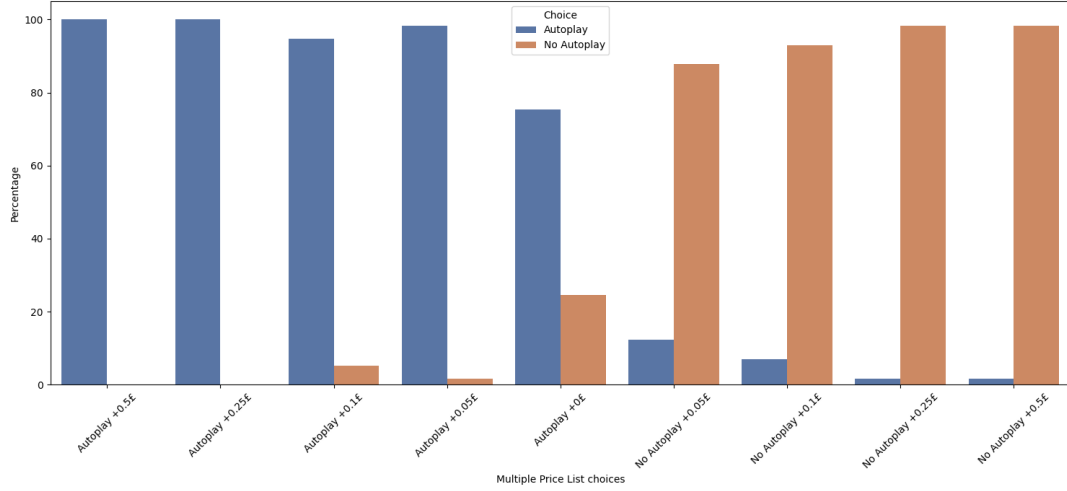


Figure 11: Distribution of Autoplay preferences in percentages

5 Discussion

There are two ways to interpret our findings. We begin our discussion first with a simplistic understanding of the results and build up for a more nuanced explanation. The straightforward conclusion based on the lack of support for Hypothesis 1 is that Autoplay does not have a distinct impact on participants' actualized decisions when contrasted to active decisions. Combined with the lack of support for Hypothesis 3 on deviations from how long participants wanted to watch videos initially and the actual preference for Autoplay in Hypothesis 4 further supports the idea that Autoplay is a 'harmless' default when it comes to media control settings and does not provoke increased consumption. Our findings allow for this conclusion, and show instead that users actually prefer autoplay to active decisions when there is money at stake. This would contradict the

presumed welfare benefits of the proposed ban in the [SMART Act](#) by the US Congress. However, this ‘simple’ interpretation of results does not consider the complexity of the issue at hand and leaves out a crucial design choice: The content of videos. To reassure, how the content was perceived by participants in our experiment was controlled by a survey question which asked them to rate the content of the videos. The median response was a 7 out of 10 rating across conditions for the content (see Table 7 for sample statistics).¹³ Yet, the selection (manual) and the presentation (randomized order) of the content are clearly different from the algorithmically curated personalized experience participants are accustomed to have on daily basis. To have a clear understanding of the implications of the experiment, it is preferable to look at the larger context in which content and design together shape the digital environment where decisions are made.

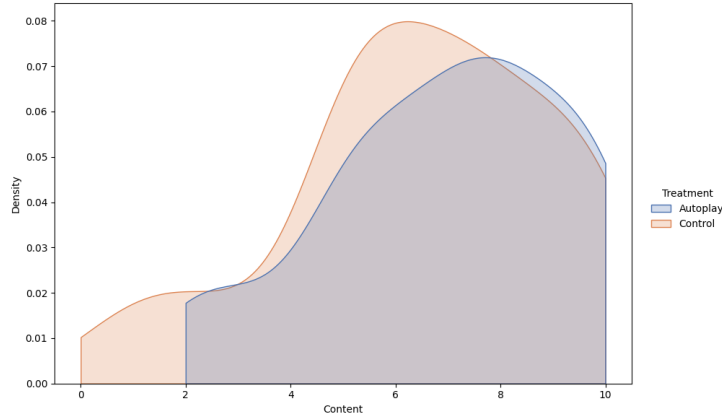


Figure 12: Density distribution of content ratings across treatments

In order to take into account this interplay between the content and the design, we rely on the insights from a recent model on media consumption ([Kleinberg et al., 2022](#)) where firms optimize over a large set of potential content and interface designs

¹³Using Shapiro-Wilk tests we cannot reject the null that distributions are significantly different from a normal distribution for both autoplay ($W = 0.932$, $p < 0.001$) and control ($W = 0.933$, $p < 0.001$) treatments. A Mann-Whitney U test between treatments cannot reject the null that distributions are significantly different from one another ($U = 6894$, $p = 0.173$), although there is some indication (see Figure 12 for the density distribution) that participants in autoplay condition gave higher ratings to the content.

parametrized by their underlying features. Consumers optimize utility for their two
 selves: The impulsive and myopic one (self 1) and the forward-looking and thoughtful
 one (self 2). Content is consumed so long as self 1 derives some positive utility, and only
 after self 1 receives negative utility can system 2 make the decision to stop consuming
 content in case system 2 was already receiving negative utility. In this model, autoplay,
 a feature that facilitates seamless content consumption, has heterogeneous effects over
 different types of content. The intuition is as follows: If the existing content already is
 such that it has a low probability that utility for impulsive self 1 is larger than zero,
 autoplay has little impact on the user behavior. In other words, on a content landscape
 with low addictiveness, autoplay would have minimal impact on engagement. Applying
 this to our setting implies that the content provided for the experiment, although itself
 curated from social media, is much less interesting for the impulsive self 1 of participants.

Further, when a user is actively engaged and desires to consume more content, the
 autoplay feature aligns with the user's intent. This supports the preference for autoplay
 found in the [MPL condition](#) regarding [Hypothesis 4](#), implying that participants were
 aware of their actual content consumption and desired to consume more content. This
 is entirely in accordance with the inference on the selection of videos not being tempting
 enough. In other words, because videos were not tempting, participants were in control
 of their consumption and could end sessions whenever their system 2 received negative
 utility from videos. In this setting, autoplay is actually helpful to participants in achiev-
 ing their desired session length as it reduces decisions costs attached to it, fully aligned
 with previous the modelization of active decisions compared to defaults by [Carroll et](#)
[al. \(2009\)](#). Putting the dual insights of the model together builds a coherent picture in
 terms of the policy implications of the present study, and we present convincing evidence
 that autoplay itself does not promote addiction in the absence of engaging content. Our
 findings go contrary to the proposed legislation which implicitly assumes all autoplaid
 content is equally addictive and welfare decreasing.

6 Limitations and Conclusion

EXPERIMENTAL DEMAND EFFECTS AND SOCIAL IMAGE CONCERNS. THE
EXPERIMENT IS SET AS A WORKING ENVIRONMENT. PARTICIPANTS PER-
CEIVED THE DECISION AS ONE TO WORK AS MUCH AS POSSIBLE.

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